XiangqiMaster — Presentation Script (spoken)

How to use this. This is what you *say*, not what you read off the slides. Lines in plain text are your words. Lines in *(italics)* are delivery cues — pauses, tone, when to click. Don't memorise word-for-word; learn the **beats** so it sounds like you, not a script. Total run time ≈ **7–8 minutes**.

Three rules to not sound stiff: (1) talk to one friendly face, then another — not the wall; (2) short sentences, breathe between them; (3) never read a bullet out loud — the slide is your poster, your voice is the story.

Slide 1 — Title

(Don't read the title. Look up, smile, then the hook. Let the first line land.)

"In 1997, a computer beat the world chess champion. Twenty years later, a completely different kind of AI beat the world champion at Go. Two games — two breakthroughs — and two totally different ideas about how a machine should *think*.

(beat)

So here's the question I kept coming back to: if you took the biggest ideas behind those breakthroughs, and set them loose on **Chinese Chess** — same board, same code, same budget — which one would actually win?

(beat, gesture to slide)

That's this project. I'm Alma, and this is **XiangqiMaster** — a fair fight between four AI paradigms on one Xiangqi board, to find out which one plays best, and *why*."

Slide 2 — What is Xiangqi?

(Lighter, quicker. Assume some of them know chess.)

"Quick ground rules, in case Xiangqi is new to you. If you know chess, you're basically 80% there. Two armies, you take turns, and you win by checkmating the enemy General — that's their king. A lot of the pieces rhyme with chess: the chariot moves like a rook, the horse like a knight.

(click / point to board)

But three things make it its own game. One — the pieces sit *on* the lines, not inside the squares. Two — there's a **river** across the middle. And three — that river changes the rules: a Soldier that crosses it suddenly can move sideways, the Elephants can *never* cross it, and the General is stuck inside his palace the whole game.

(beat)

So: familiar enough to follow, different enough to be its own puzzle."

Slide 3 — Similar game, harder problem

"And here's the thing — those quirks don't just add flavour, they make it genuinely *harder* for a computer. Two examples. The cannon can only capture by **jumping over** another piece — so every

capture depends on what's in the way. And the horse can be physically **blocked** by a piece right next to it.

(beat)

Little rules — but they blow up the number of things the AI has to think about. Xiangqi has a bigger branching factor than chess and longer games. Same family as chess — meaner problem. Which, for testing AI, is exactly what you want."

Slide 4 — Xiangqi is a serious AI benchmark

(Confident. This is where you earn academic credibility — then open the gap.)

"And I'm not the only one who thinks so. Researchers recently used Xiangqi to test whether large language models can *actually reason* — because this game forces real, multi-step thinking, not pattern-matching. There's a paper in TMLR, called JiangJun, that trained on over ten thousand human games just to study one way to play it well. And there's an AlphaZero-style player from Yilmaz and colleagues.

(beat — slow down, this is the pivot)

So the game is taken seriously. But notice something: every one of these studies looks at **one** method. One team does imitation. Another does AlphaZero. Nobody lines them up on the *same* board and asks the obvious question —

(beat)

— which approach actually wins? That's the gap I'm filling."

Slide 5 & 6 — The four paradigms

(This is the heart of the talk. Give each one ONE clear sentence + its famous cousin. Use your hands — count them off, one to four. Don't rush.)

"So I built four players, each using a different big idea in AI. Think of them as going from 'pure calculation' to 'learned intuition'.

(1) **Minimax.** Old-school brute-force search — no learning at all. It just looks ahead and assumes both sides play perfectly. This is the Deep Blue idea, the one that beat Kasparov.

(2) **Imitation Learning.** This one learns by **copying humans** — I showed it millions of human moves and it learned to play like us. This is how AlphaGo first learned Go.

(3) **Self-Play Reinforcement Learning.** No human data at all — it plays against itself millions of times, wins and loses, and slowly figures out what works. This is the family behind OpenAI's Dota bots and the training behind ChatGPT.

(4) **Monte Carlo Tree Search with a neural network** — the AlphaZero recipe. The network gives it *intuition*, and then it *searches* to double-check that intuition before every move.

(beat)

Plus two reference points: a random player as the floor, and Pikafish — a superhuman engine — as the ceiling. Everybody plays everybody. One league, one scale."

Slide 7 — What I'm researching + objective

"So, concretely, I'm chasing three questions.

One — **the ranking**: on a single Elo scale, how do the four actually stack up?

Two — **the imitation effect**: if I let a model learn from humans first and *then* do self-play, does that help it, or hold it back? That turned out to be the most surprising part — I'll get to it.

Three — **what limits strength**: when an agent hits a wall, is it the search, the network size, the data, or the algorithm itself?

(beat)

And the goal isn't just a number. It's a **controlled, reproducible answer** to 'which paradigm wins and why' — plus a reusable evaluation league that the next student can pick up and build on."

Slide 8 — What's already built

(Tone shift: proud, grounded. This is where you show it's real, not a proposal.)

"And this isn't a plan — it's already running. I built the whole thing from the ground up: the game engine with every Xiangqi rule, the data pipeline, all four agents, three different training systems, the Elo tournament, and a live demo.

(beat)

Just to give you a sense of scale — the engine, the 2,000-move action space, over eleven million training positions, and 224 automated tests keeping it honest. The foundation is done. What's left is scaling it up — and I'll show you exactly what and when."

Slide 9 — Results so far

(Slow down. This is your payoff slide. Point at the chart, then deliver the twist.)

"Here's what came out. On the Elo scale, the **MCTS + network agent won** — the AlphaZero-style one. No huge surprise there; intuition plus search is a strong combo.

(beat — now the interesting part)

But three things *did* surprise me. First: imitation learning, just copying humans, **beat every self-play agent**. Learning from scratch on one laptop simply wasn't enough. Second: I tried using the superhuman engine as a teacher — and even *that* didn't beat plain human imitation. And third — this is the big one —

(beat)

I tried four different ways to push the agents stronger: more search, more training, a stronger teacher, a bigger network. And all four hit the **same ceiling**, around 1690 Elo.

(beat)

That tells me something important. On this scale, the thing holding the AI back isn't the *algorithm* — it's the **capacity and the compute**. And knowing *that* is exactly what tells me where to push next."

Slide 10 — Live demo

(Switch to the web app. Keep talking while it loads — never stare at a loading screen in silence. Run a FAST matchup: Pikafish MAX vs Minimax d1.)

"Let me actually show you. This is the live demo — I pick two of my agents, hit play, and we watch them fight it out.

(click Play)

Now, watch the graph at the bottom, because that's the whole story in one picture. When the red area grows, Red is winning; blue means Black is winning. You can literally see the moment one side takes over.

(let a few moves play)

And there it is — one side pulls ahead, the lead climbs, checkmate. That graph is how I turn 'which AI is better' into something you can actually *see*, move by move."

Slide 11 — What's not done + schedule

(Honest and confident — examiners respect knowing your own gaps.)

"So what's left? The foundation is solid; now it's about **scale**. Next, I take those experiments bigger — larger networks, more self-play, more compute — to see if I can break that 1690 ceiling. Then I calibrate everyone against the engine on one absolute scale. Then I write it all up.

(gesture to timeline)

That's the road to my defense at the end of May 2027. Clear steps, real deadlines — I know exactly where this is going."

Slide 12 — Thank you

(Land it. Don't trail off. Look up, pause, then the closer.)

"So — four AI paradigms, one Xiangqi board, one fair fight. I already know which one wins today, and I know exactly what's holding the rest back. The next six months are about pushing past that ceiling.

(beat, smile)

Thank you — I'd love to take your questions."

Quick delivery cheat-sheet

- **The hook (Slide 1)** and **the twist (Slide 9)** are the two moments that make or

break the talk. Rehearse those until they feel effortless.

- **Pause after every big number and every surprise.** Silence makes you sound confident, not nervous.
- **Demo = pick a fast matchup** (Pikafish MAX vs Minimax d1) so it finishes in a couple of seconds. Keep narrating while it loads.
- **If you blank:** your spine is always the same four words — *engine, agents, results, what's next.* Find your place and keep going.
- **Phrases to avoid** (they sound canned): "In this presentation I will…", "Furthermore", "As you can see", "In conclusion". Just say the thing.